

2D Brain Age Estimation and Alzheimer's disease Classification using Deep Learning

Emil Marcus Buchberg, Frederik Hecter Kowalski, Kent Vugs Nielsen

Abstract—In recent years, deep learning has emerged as a valuable tool for estimating brain age. Neuroimaging is crucial in capturing age-related brain changes and serves as a valuable biomarker for the resurgence of neurological conditions. To detect individuals with Alzheimer's disease, we propose an extension to the established Global Local Transformer method that first extracts deep features from T1-weighted images. These features are then employed in a simple classification multi-layer perceptron to distinguish subjects as normal aging and cognitively healthy or age-diverging and suffering from Alzheimer's disease.

A model selection process determined the optimal configuration for the Global-Local Transformer of 6 local patches, 5 2D slices per subject, and 6 transformer blocks. Experimental findings reveal that the global-local transformer yields a mean absolute error of 5.34 years when applied to healthy subjects. However, a slight elevation of 0.74 years in the mean absolute error is observed for individuals diagnosed with Alzheimer's disease, indicating a relatively limited differentiation in brain age between these two subject groups. Despite the observed brain age difference, the classification multi-layer perceptron attains a commendable accuracy of 76.91%, comparable to contemporary state-of-the-art techniques.

I. INTRODUCTION

AGING is a biological phenomenon characterized by a gradual manifestation of physiological alterations, leading to a progressive decline in functional capacity. The process of aging exhibits individual variability with regards to its intensity and pace due to genetic and environmental factors [1], [2] [3], [4]. Nor, in fact, does the aging process appear to be uniform within an individual. Considerable variation exists in age-related diseases and the overall lifespan of individuals. This variability has motivated biogerontological research endeavors aimed at characterizing aging from a biological standpoint to identify aging biomarkers that surpass chronological age in their capacity to predict disease susceptibility and residual lifespan of an individual [4]. Likewise, brain changes do not transpire uniformly across all brain regions. With advancing brain age, there is a tendency towards cognitive decline and heightened susceptibility to neurodegenerative diseases and dementia. Furthermore, chronic neuropsychiatric disorders may be exacerbated by the aging process. Brain age is defined as the biological age of an individual's brain and has proven to be a valuable biomarker for neurological and psychiatric conditions [3], [4], [5].

Neuroimaging methods, such as magnetic resonance imaging (MRI), offer potent mechanisms for capturing brain changes that transpire with aging. Specifically, such techniques have proven to be valuable tools for probing the impact of age-related alterations in brain structure and function, encompassing morphometric atrophy and connectivity reduction.

Multimodal MRI, involving structural MRI (sMRI), functional MRI (fMRI), and diffusion tensor imaging (DTI), can capture diverse structural and functional features of the human brain in a non-invasive manner. As a result, multimodal MRI has become increasingly prevalent in research endeavors targeting the brain aging process, as well as in the clinical diagnosis of brain disorders [1], [3], [4].

Machine learning techniques and deep learning models can be employed to estimate brain age by analyzing neuroimaging data [2], [3], [4] [6]. The difference between estimated brain age and chronological age is referred to as the estimated age difference (EAD), denoting the degree of divergence from the standard trajectory of brain aging. The MRI-derived brain age exhibits associations with brain health, with a positive EAD reflecting an accelerated aging process and a negative EAD indicating decelerated or healthy aging. A positive EAD is further linked to an elevated risk of cognitive decline and neurodegenerative disorders, such as Alzheimer's disease (AD) and mild cognitive impairment (MCI), which represents the transitional stage from the healthy state [2], [3], [6]. No definitive cure exists for AD, emphasizing the significance of early diagnosis for appropriate treatment. Precise estimation of brain age is instrumental in enabling early detection of AD [2], [3], [6]. Brain age estimation involves leveraging the healthy brain to identify age-related variations, with age regression models developed using features derived from healthy brain images. Through such models, an individual brain image's complex multidimensional aging pattern is reduced to a single value referred to as "estimated brain age". Deviation from the chronological age can aid in diagnosing various neurological diseases and brain conditions, thereby helping to identify accelerated aging and brain atrophy. Thus, brain age estimation involves training a machine learning model on a healthy aging population, thereby enabling the extraction of age-specific patterns learned from healthy brain MRIs to estimate brain age on unseen data [2], [6].

The typical framework for brain age estimation involves using MR images as a data source, followed by data pre-processing, feature extraction, and age regression [2], [3]. Convolutional neural networks (CNNs) have demonstrated remarkable performance in various applications as part of frameworks estimating brain age. CNNs can extract deep features on the whole input image and local patches segmented from the input image, thus enabling a more comprehensive analysis of brain structures [6]. We present an extension of an existing approach, improving the performance of MRI analysis by leveraging a patch-based strategy to capture fine-grained information. To effectively integrate the local patch context,

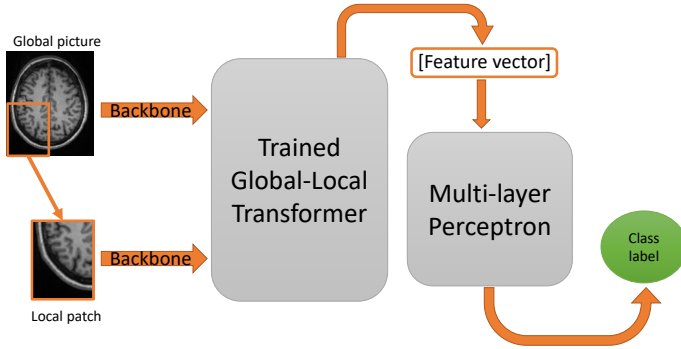


Fig. 1: Overview of proposed method

we employ the Global-Local Transformer (GLT) proposed by He et al. [6]. As part of the GLT, we utilize a CNN for deep feature extraction and their proposed transformer for regression analysis to estimate brain age. A learned feature vector is extracted and used in a multi-layer perceptron (MLP) to classify AD. Our proposed approach provides a promising framework for accurate diagnosis of AD and improving MRI analysis.

II. RELATED WORK

In recent years, Deep Learning (DL) has gained prominence as a novel Machine Learning (ML) methodology in neuroimaging. CNNs have become widely used among DL architectures and have accumulated significant attention for their success in image analysis. DL differs from conventional ML in that it can automatically abstract low-to-high-level latent feature representations. This ability allows for more efficient and accurate analysis of medical imaging data [8]. Table I summarizes related studies in the literature for brain age estimation using CNNs. The studies conducted by Cole et al. [9], Cai et al. [3], He et al. [6], Valliani et al. [10], and Senanayake et al. [11] utilized a diverse dataset obtained from multiple sources, commonly including the Alzheimer’s Disease Neuroimaging Initiative (ADNI) [12] dataset. The OASIS-3 dataset was utilized in the studies conducted by Cole et al. [9] and He et al. [6], serving as one of the datasets employed in the analysis. The OASIS-3 dataset is further described in III-A.

In [9], a 3D neural network architecture consisting of five repeated blocks was proposed. Each block included a $(3 \times 3 \times 3)$ convolutional layer, followed by a rectified linear unit (ReLU), another $(3 \times 3 \times 3)$ convolutional layer, a 3D batch-normalization layer, another ReLU, and a $(2 \times 2 \times 2)$ max-pooling layer. The number of feature channels was set to eight in the first block and doubled after each max-pooling layer. The final age prediction was obtained using a fully connected layer, which mapped the output of the last block to a single output value. The model achieved a mean absolute error (MAE) of 4.16 years in the age range of 18-90. In [3], a framework for brain age estimation using graph transformer geometric deep learning was proposed. The framework utilizes sMRI and DTI data to construct and combine multimodal brain networks. It consists of a convolutional autoencoder for

feature extraction, multi-level construction of brain networks in a graph structure, and a graph transformer network to model the multimodal brain networks for brain age estimation. The model achieved a mean absolute error (MAE) of 2.71 years in the age range of 46-81. In [6], a VGGNet-based CNN, also seen as the backbone in Figure 2, is employed to extract deep global context features from a global pathway and deep local features from the local pathway from the input image to produce local feature vectors fused with global context from the input images. The feature vectors are used for brain age estimation. The backbone is composed of eight blocks, each containing a convolutional layer with a kernel size of (3×3) , a batch normalization layer, and a ReLU activation layer. A max-pooling layer with a kernel size of (2×2) is applied after every two blocks to gradually reduce the spatial dimensions. The backbone transforms the input image into a deep feature, which represents the abstract and high-level features of the input image. The final layer for brain age estimation is a fully-connected layer that maps the feature vector (obtained after an average pooling layer) with 512 dimensions to the predicted brain age. The GLT outputs predictions equal to the amount of local patches extracted from an image, and one for the global pathway. He et al. [6] extracts 8 local patches in their trained model, which grants nine predictions. The final prediction is an average of all predictions. The model achieved a mean absolute error (MAE) of 2.70 years in the age range of 0-97.

TABLE I: SUMMARY OF STUDIES FOR BRAIN AGE ESTIMATION USING REGRESSION

Study	Modality	#Subjects	Age ranges	MAE (years)
Cole et al. [9]	sMRI	2001	18-90	4.16
Cai et al. [3]	sMRI	16458	46-81	3.09 ± 0.05
Cai et al. [3]	sMRI + DTI	16458	46-81	2.71 ± 0.07
He et al. [6]	sMRI	8379	0-97	2.70 ± 0.03
Adopted GLT	sMRI	818	42-97	5.34 ± 4.72

In addition, certain studies have incorporated experiments aiming to transfer a pre-trained deep learning (DL) model onto a novel dataset in order to investigate its classification abilities on Alzheimer’s disease (AD) and normal controls (NCs). Table II summarizes relevant studies in the literature on transfer learning and AD classification. The studies indicate that the estimated age difference (EAD) increases as subjects transition from NC to MCI to AD, implying that it is a valuable biomarker for evaluating dementia severity and aiding early diagnosis of AD. [3].

In [10], the authors propose a fusion architecture that can merge data from two modalities: 3D MR volumes and neuropsychological measure-based feature vectors. The challenge is to match the dimensions since the former has millions of features while the latter has only 35. They suggest a pipeline that meaningfully reduces the dimensions of 3D MR volumes to a comparable dimension so the two feature vectors can be merged. The pipeline includes dilated convolution, residual connections, dense connections, and (1×1) convolutions. Dilated convolution is used to incorporate global-context into the convolution operation, while residual and dense connections

allow easier training of deeper networks. (1 x 1) convolutions are employed for dimensionality reduction in the feature space while maintaining spatial relevancy. The model achieved an accuracy of 76.0%. The study by [11] proposed a deep residual network for MRI image analysis. The architecture consisted of 18 layers and was based on the ResNet-18 model. To improve the model’s performance for the specific task of MRI image analysis, the convolutional layers were used as filters, and two fully-connected layers were added. The weights used for the convolutional layers were pre-trained on the ImageNet dataset. The entire network was fine-tuned on MRI images to adjust the model’s parameters and improve its accuracy for the task. The model achieved an accuracy of 81.3%.

We propose a framework that combines the two presented domains of regression and classification for NC and AD subjects presented. (1) We implement a unique data pre-processing pipeline to ensure data homogeneity and quality. (2) We employ the GLT, as proposed by He et al. [6], to conduct regression modeling of MRI scans. (3) We extract the global-local context feature vector from a GLT network trained in a regression setup because it contains pixel-level morphological information of the brain, and employ it in a classification task with a simple MLP. (4) We address the issues of data leakage and imbalance, leading to improved generalization and reduced overfitting. A complete overview of the proposed model is seen in Figure 1.

Our model achieves a mean absolute error (MAE) of 5.34 years over a lifespan range of 42-97 years, making our error competitive and comparable to that reported in [6] for the same age group. In addition, the MLP achieves an accuracy of 76.9%.

TABLE II: SUMMARY OF STUDIES FOR ALZHEIMER’S DISEASE (AD) CLASSIFICATION

Study	AD vs. NC
Cai et al.* [3]	ACCURACY=85.9%
Valliani et al. [10]	ACCURACY=81.3%
Senanayake et al. [11]	ACCURACY=76.0%
Proposed Method	ACCURACY=76.9%

* Has data leakage in their results.

III. MATERIALS

Herein, we present the utilized data and the pre-processing techniques applied. We also provide a detailed overview of its applications in regression and classification tasks.

A. Dataset

The OASIS-3 dataset [13] is used for this project. OASIS-3 is a longitudinal study where subjects are studied over a prolonged period, and therefore most subjects are featured in multiple MRI sessions. OASIS-3 totals 2842 MRI sessions across 1377 subjects, with each session including T1-weighted images (T1WIs). A T1WI is a commonly used modality for brain age estimation because it offers high spatial resolution and good tissue contrast enabling accurate analysis of morphological changes related to aging [1] [4].

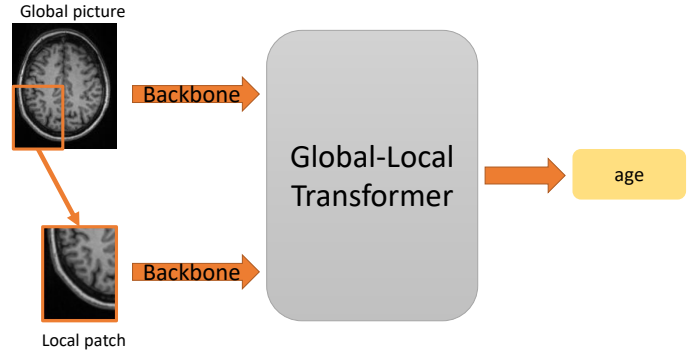


Fig. 2: Overview of the GLT framework. We exclude the global prediction, due to its inaccuracy compared to the local predictions. We still utilize the global pathway to infuse the local patches with global-context.

We downloaded the OASIS-3 dataset from Xnat.org using a script from [16]. The script downloads and organizes the files into the Brain Imaging Data Structure (BIDS) [17], which is a prerequisite for using the pre-processing pipelines in III-B. Some subjects were discarded due to missing labels or failing the quality check described in III-B. It was necessary to perform additional data preparation due to structural and file naming discrepancies incompatible with the BIDS. A Python script was employed to rectify these issues and make the necessary corrections [18].

Of the 1377 subjects, all within the age range 42-97, 755 are cognitively healthy adults, and 622 are experiencing cognitive decline at various stages. The framework of the proposed model requires two groups; an NC subject group and an AD subject group. The NC group exclusively comprises subjects with no cognitive impairments or other neurologically-related diseases. Subjects exhibiting cognitive decline, who fall into categories such as mild cognitive impairment (MCI) or suffer from non-AD, are excluded, meaning that subjects with various AD types are the only ones comprising the AD group. The resulting frequencies of the respective groups are highlighted in Table III.

Strict legislation usually applies when handling health data due to the inclusion of sensitive data acquired from patients. However, the General Data Protection Regulation (GDPR), recital 26 states: “*The principles of data protection should therefore not apply to anonymous information, namely information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous in such a manner that the data subject is not or no longer identifiable*” [19]. The OASIS-3 dataset has been subject to rigorous anonymization measures to protect the subjects’ privacy. In compliance with the data usage agreement, we strictly refrain from any attempts to ascertain the identities of the subjects included in the dataset. However, we can reasonably assume that the implemented anonymization measures effectively mitigate potential privacy risks such as linkage and de-anonymization attacks.

TABLE III: OASIS-3 DISTRIBUTION OF SESSIONS WITH HEALTHY AND COGNITIVELY DECLINING SUBJECTS

	#Subjects	#Sessions	Age statistics
NC	663	1116	67.96 $\pm$ 9.47
AD	155	179	75.62 $\pm$ 7.72

B. Image Pre-processing

It is imperative to ensure the homogeneity and adequate quality of the acquired T1WIs for the purpose of conducting group-level analysis. Training a DL model with T1WIs inadequately registered to a template or containing artifacts ultimately harm the model’s performance. Therefore, the T1WIs undergo a meticulous pre-processing process which can be broken down into five steps: bias field correction, image registration and normalization, pre-processing quality check, slice extraction, and slice-to-tensor conversion. All the applied pre-processing tools originate from the Clinica and ClinicaDL libraries.

The T1-linear pipeline from Clinica [20] [21] has three components. First, bias field correction using the N4ITK method [22] to correct low-frequency intensity non-uniformity [23]. In other terms, N4ITK corrects the voxel intensity values affected by the spatially varying intensity bias. Intensity values refer to the grayscale values of each voxel ranging from 0 (black) to 255 (white), with the in-between intensities representing various shades of gray. A bias field within a specific region causes the voxels in that region to deviate from their true values. Consequently, the same tissue is mapped to different intensity values throughout the image. Various factors contribute to a bias field, one of which is the presence of acquisition artifacts which, for example, can be caused by subject motion during the scanning process. Thus, bias field correction is necessary to preserve the spatial invariance of the processed image [24].

The second component is an affine registration process using the SyN algorithm [25] from ANTs [26] to align each image to the MNI space with the ICBM 2009c nonlinear symmetric template [27]. The registration process aligns brains of different sizes, shapes, and dimensions such that an arbitrary voxel corresponds to the same anatomical location of the brain across every subject, an essential feature to performing group-level analysis.

In the third and final component, the images are cropped to remove unnecessary zero-pixels surrounding the skull. The resulting images are of size 169x208x179, with (1mm x 1mm x 1mm) voxels.

The quality of the pre-processing step is checked with the quality checking pipeline [29] [31] offered in the ClinicaDL library. It returns an overview of the input images and their respective pass probabilities. A 3D image receiving a low pass probability indicates a poorly registered image. A review showcased that images with a probability below 0.8 were severely distorted, and all were discarded.

The proposed model takes 2D slices as inputs. We use the pipeline in [28] from ClinicaDL [29] [30] to transform the 3D image outputs to 2D slices. The pipeline extracts 2D slices

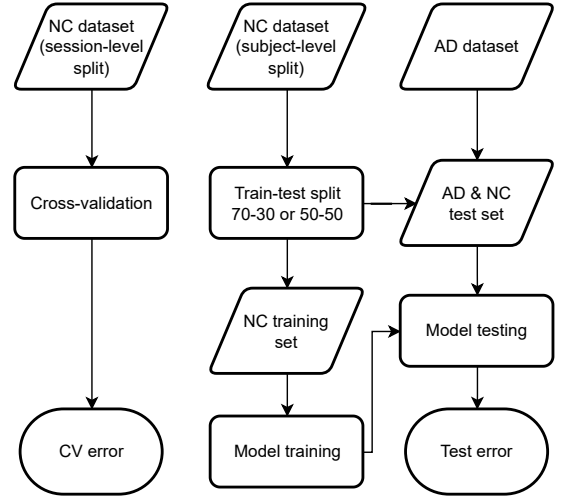


Fig. 3: Flowchart of data utilization

from the center of the pre-processed 3D images and returns them in a tensor format. The pipeline has two key parameters: slice direction and number of slices. One can extract up to 187 slices from a single 3D image, and the optimal value is estimated in V-C.

C. Validation & Testing

The data utilization for adjusting the model configuration and testing generality is twofold. Figure 3 provides an overview of the overall process.

We first address issues concerning data leakage in longitudinal studies. As the OASIS-3 dataset is a longitudinal study, data leakage can occur in various ways, and it is essential to address and mitigate it in order for us to report credible results. One potential source of data leakage is the propagation of sessions, from the same subject, into multiple sets among the training, validation, and test set. Utilizing multiple 2D slices per subject introduces a potential threat of data leakage. We avert from shuffling before the initial splits during training and validation to prevent 2D slices from the same subject from propagating into different sets. However, the individual sets are shuffled once the training sets and folds have been established. Bäckström et al. [15] reported an accuracy gain of 8 percentage points when utilizing a session-level split that suffered from data leakage compared to a subject-level split without data leakage. Thus, data leakage causes biased data splits and overly optimistic results [14].

Looking at Figure 3, we employ five-fold cross-validation (FFCV) during the model selection phase to identify the optimal parameters and hyperparameters of the GLT. Note that FFCV is done on the NC dataset using a session-level split. We acknowledge that during our model selection experiments, we were unaware of the potential pitfalls of longitudinal studies and data leakage. Therefore, the model selection process presented in subsection V-C suffers from data leakage, as sessions from the same subjects were present in both the training and validation set. Consequently, the reported results are over-optimistic and possibly misleading, undermining the model

selection's credibility. Unfortunately, we could not rectify this error due to limited resources and the time-consuming nature of the model selection process. Nevertheless, we maintain that there is some degree of credibility in the results when comparing the performances of different parameter configurations, primarily due to the utilization of the same data and FFCV folds for all reported configuration errors.

After the model selection process, the reported data leakage was corrected. Consequently, a limit of one session per subject was imposed, discarding several sessions in the process. Hereafter, the GLT is trained on either 50% or 70% of the corrected NC data. The remaining 50% or 30% of the NC group is then combined with all AD subjects to form a mixed dataset. The pre-trained GLT model is transferred to the mixed dataset and cross-validated using stratified FFCV in conjunction with the MLP. Data leakage can arise during transfer learning if the source and target datasets overlap [14]. In our specific setup, this would occur if data from the initial training data used for the GLT model leaked into the mixed dataset used for cross-validating the MLP model. The cross-validation errors are then used for reporting the proposed model's performance.

IV. METHODS

In this paper, we adopt and employ two different methods that, in conjunction, accomplish the task of separating NC subjects from AD subjects by classification. Therefore, this paper's main area of interest is the attention mechanism and transformer proposed by He et al. [6], used to retain the deep feature brain information initially designed for regression modeling. We consequently seek to provide an explanation of the global-local self-attention mechanism, which is the method that infuses the global context with deep local features, and the global-local transformer, but not the entire architecture of He et al.'s [6] framework. Some of the backbone is shortly covered in I, and the entirety of the framework's architecture is fully explained in the original paper [6]. The second method, a classification neural network, constitutes our expansion to the proposed model, built on top of the GLT architecture. Similar to Cai et al.'s [3] approach, we extract a learned feature vector that serves as an input to a classification MLP from which a final classification is produced. We introduce the elementary theory underlying the simple MLP and then present the more advanced theory connected with the GLT.

A. Classifier

The MLP classifies a subject based on a binary class label where 0 (negative) represents the NC group and 1 (positive) represents the AD group. The MLP is a simple classification network composed of layers with ReLU and Sigmoid neurons. The architecture consists of an input layer, two hidden layers, and an output layer as depicted in Figure 4. As one patch in GLT yields a feature vector of size 512, the size of the input layer is given as $n \times 512$ with n individual patches extracted from a single 2D slice. The number of patches is a parameter in GLT that we later find through experiments. Both hidden layers retain the same size of twice the size of the input layer.

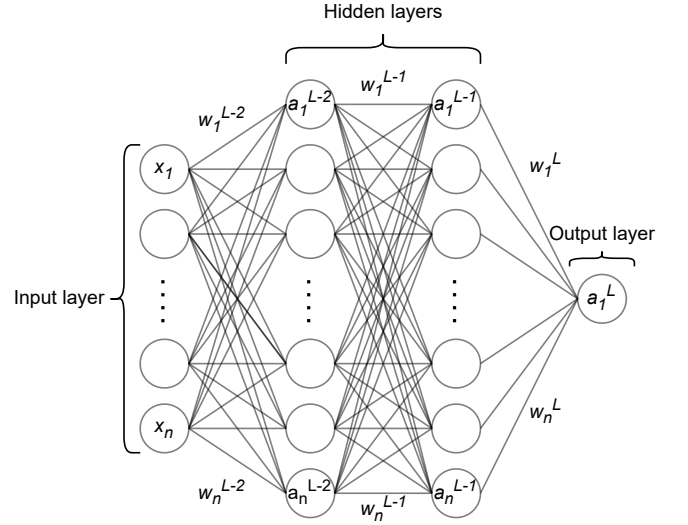


Fig. 4: The MLP architecture with an input layer of n neurons, two hidden layers with $n \times 2$ neurons, and an output layer with a single output neuron.

The neurons are compressed into a single output neuron from the final hidden layer to the output layer.

Forward propagation is the process of computing intermediate outputs and a final output in a neural network. $z^l = w^l \cdot a^{l-1} + b^l$ describes that each output, z , is a linear function of the total input from the previous layer involving learned weights w^l and biases b^l for every input-output pair to a neuron [32]. Thus, z^l represents a weighted sum of the total input from neurons in the $l - 1$ th layer. Since two adjacent linear layers yield a new linear function, one should always include an activation function in-between linear layers. Moreover, an activation function introduces non-linearity into the model, allowing it to model complex non-linear relationships between inputs and outputs. For our case, we use two activation layers applying the ReLU function and a single activation layer applying the Sigmoid activation function (SAF). In a binary classification task, it is typical for the final output to employ the SAF. The SAF is given as

$$\text{Sigmoid}(z^l) = \sigma(z^l) = \frac{1}{(1 + e^{-z})},$$

maps its inputs to a probability range between 0 and 1 [32]. Subsequently, the final output neuron can be rounded to the nearest (0, 1) using the following rule:

$$\text{output} = \begin{cases} 1 & \sigma(z^L) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

This process yields the final prediction of the MLP and concludes the forward propagation process.

The MLP employs backpropagation to adjust the weights and biases. A process involving two passes through the network. During the first pass, forward propagation is used to compute the output. In the second pass, the network is traversed in reverse, from the output to the input layer, allowing for weight and bias updates. By iteratively optimizing the values of the weights and biases with respect to a chosen

cost function, the MLP learns the optimal configuration that minimizes the cost function as seen in Algorithm IV-A formulated by [M. A. Nielsen] [33]. In practice, the algorithm includes an another outer loop iterating through epochs. Regarding notation, the neuron indices at layer basis, as seen in Figure 4, are dropped and presented in vector form. In addition, σ was earlier used to denote the SAF but here denotes an arbitrary activation function.

Backpropagation algorithm

- 1) **Input:** training example x
- 2) **Forward propagation:** for each layer $l = 2, 3, \dots, L$ compute:

$$z^l = w^l a^{l-1} + b^l \text{ and } a^l = \sigma(z^l) \quad (1)$$

- 3) **Output error** δ^L : compute the vector

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (2)$$

- 4) **Backpropagate the error:** for each layer $l = L-1, L-2, \dots, 2$ compute

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (3)$$

- 5) **Output:** the gradient of the cost function is given by

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \text{ and } \frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (4)$$

Starting from 2) in the backpropagation algorithm, a training example is propagated through the network to produce the output activations. In 3), the corresponding error in the output layer is calculated. $\nabla_a C$ is a vector of gradients, $\frac{\partial C}{\partial a_i^L}$, of the cost function C with respect to the output activations a which are element-wise multiplied with the derivative of the activation function at the output layer. $\sigma'(z^L)$ represents the sensitivity of the output activations to changes in the weighted sum. Step 4) illustrates the computation of δ^l and the backpropagation of the error from $(l+1)$ th layer back through the network by utilizing the chain rule. Thus, combining Equation 2 and Equation 3 allows us to compute δ^l for any layer in the network. The final step 5) gives the gradient of the cost function with respect to the weights $\frac{\partial C}{\partial w_{jk}^l}$ and biases $\frac{\partial C}{\partial b_j^l}$ of each layer. The gradients are subsequently used to update the weights and biases to reduce the cost.

B. Self and multi-head attention

The global-local attention mechanism shares foundations with the well-known self-attention method, used by Vaswini et al. [7] as the primary mechanism of their transformer architecture. Self-attention is distinguished from regular attention by its property to focus on a single sequence, which allows the model to capture dependencies within a single input sequence instead of associating an input and output sequence. Similarly to Vaswini et al. [7], the attention mechanism we utilize is a scaled dot product attention, which allows for relevance computation of positions in an input sequence. Attention comprises “queries” and “keys” of dimension d_k and “values” of dimension d_v . Realistically we want to compute a set of

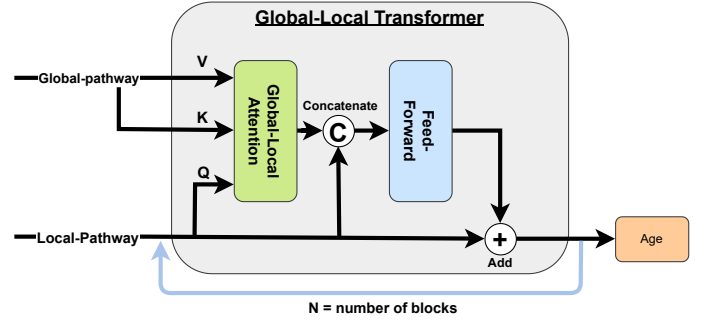


Fig. 5: The Global Local Transformer, proposed by He et al. [6]. The global and local pathways extracts features from the same T1WI. We exclude the global prediction.

“queries”, “keys” and “values” concurrently, and so we pack the “queries” into matrix Q , the “keys” into matrix K , and the “values” into matrix V . We then compute the similarities of Q and K , by computing the dot product, and divide them by $\sqrt{d_k}$ to scale them. A softmax function is applied to normalize the computed similarities and obtain what we name the attention scores, which are the weights to the values in V [7]. The attention is expressed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (5)$$

By computing the attention, we compute a representation of the input sequence. Since the matrices Q and K are different matrices, it allows for asymmetry, a key factor in the global-local attention mechanism.

Vaswini et al. [7] also proposes using multi-head attention. The purpose of multi-head attention is to capture patterns and dependencies that might not be learned by a single attention mechanism. We split and linearly project the “queries”, “keys” and “values” h times utilizing different learned linear projections into dimensions d_k , d_k , and d_v , and subsequently perform the attention function in each projection concurrently and obtain d_v dimensional output values [7]. h is the count of attention-heads. Multi-head attention is presented as

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \quad (6)$$

where

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V).$$

The matrices W_i^Q , W_i^K , and W_i^V are projection matrices belonging to head i . The W^O matrix is the final projection matrix, which is the whole multi-headed attention head.

C. Global-local attention

The global-local attention mechanism proposed by He et al. [6] differs from the traditional scaled dot product attention in terms of the construction of “queries” “keys” and “values”. The “query” is from the local feature f_l extracted from the local pathway. The “keys” and “values” are from the global feature f_g extracted from the global pathway. The deep feature f_g is extracted from an input T1WI with dimensionality $f_g^{h \times w \times d}$ at a pixel level with height h , width w , and number

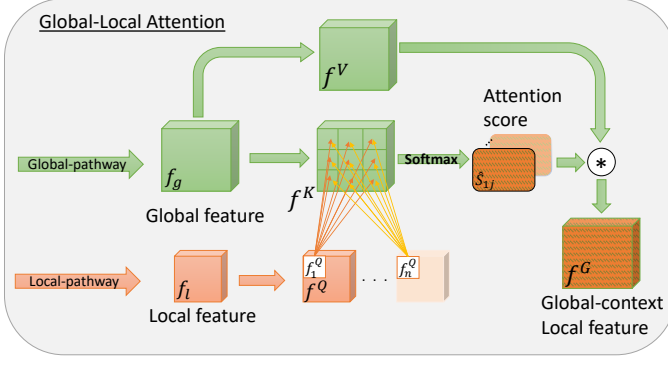


Fig. 6: The Global-Local attention mechanism, details the attention block from Figure 5. The output is the Global-context Local feature.

of channels d . Whereas the deep feature f_l is extracted from a patch on the same T1WI as f_g , and therefore has dimensionality $f_l^{h' \times w' \times d}$, where $h' < h$ and $w' < w$. The deep local feature f_l contains information regarding the local extracted patch details, and the deep global feature f_g contains the contextual information of the entire image. To fuse the local patch information with the global contextual information, f_l is projected into the “query” space f^Q , and f_g into “key” space f^K and “value” space f^V . Since f^Q is the “query”, we calculate the similarity of its position on each pixel level i in f_i^Q to all positions in f_j^K to obtain the similarity function $s_{ij} = f_i^Q \cdot f_j^K$ [6]. We realize we have obtained the dot product on a pixel level from Equation 5, and so we scale it with our dimensionality $\sqrt{d}$, and apply the softmax function to normalize it, which derives the function

$$\hat{s}_{ij} = \text{softmax} \left(\frac{s_{ij}}{\sqrt{d}} \right). \quad (7)$$

The computed result from Equation 7 is the attention scores, which are the weights of the “values” from the global pathway. Subsequently, to obtain the global-context for each pixel i at position f_i^Q , we sum the weights with the values from the global pathway $f_i^G = \sum_j \hat{s}_{ij} \cdot f_j^V$. As in Equation 5, we are interested in doing large sets of computations simultaneously; we want to compute every pixel i in f_i^Q . To handle computations for all pixels in f^Q concurrently, we organize the “queries” for each pixel into a matrix F^Q with dimensions $h' \times w' \times d$. Similarly, f_j^K and f_j^V is reshaped into F^K and F^V , with size $h \times w \times d$, and comparably to Equation 5, we obtain:

$$F^G = \text{softmax} \left(\frac{F^Q (F^K)^T}{\sqrt{d}} \right) F^V, \quad (8)$$

which is the global-local attention that we call the global-context local feature. We synonymously use F^G for the global-context local feature. Because F^G results from the asymmetry between F^Q and F^K , the size of F^G is the equivalent size of F^Q . The GLT method proposed by He et al. [6] additionally

employs multi-head attention. The number of attention heads h is set to 8, and the local and global features are split on the channel dimension. As seen in Equation 6, the global-local attention function is applied to every head. The outputs are concatenated into one final feature, which is the same size as the feature inputs [6].

D. Global-local transformer

NN Transformers have been successfully implemented due to their ability to handle long-range dependencies by employing the aforementioned self-attention [7]. The transformer architecture was proposed with the self-attention by Vaswini et al. [7] and is currently considered a state-of-the-art method. He et al. [6] employs parts of the transformer architecture to further fuse the global contextual information with the local details N times. We consider $N = 1$ to be a transformer block, which subsequently leads to having N identical transformer blocks in practice to iteratively fuse the global-context with local features. The architecture can be observed on Figure 5.

The output of the attention block illustrated on Figure 6, F^G , contains the global contextual information from the global pathway, and we, therefore, concatenate it with the local feature from the local pathway, as observed in Figure 5. We further fuse the two features in a feed-forward block. The feed-forward block is constructed by two convolutional layers that contain 512 entries and kernel size (1×1) , with a batch normalization layer and ReLU activation functions [6]. The feed-forward layer’s output is added to the local feature, and subsequently, the local feature constitutes the local details with the contextual information. This process can be repeated N times, where the output of a transformer is yet again fed into an identical transformer to infuse the local feature with the global contextual information iteratively, which constitutes the Global-Local Transformer [6].

V. REGRESSION EXPERIMENTS AND RESULTS

In this section, we present the optimal configuration for the adopted GLT model, the first of the two conjunctive methods used to separate NC and AD subjects. Finally, we present the outcomes of our experiments with the configured GLT network and provide an interpretation of the results.

A. Network Training

The network is trained and implemented using the PyTorch library [30], where we use the Adam optimizer. The mean absolute error (MAE) is chosen as the loss function, which is defined as

$$\text{loss} = \frac{1}{n} \sum_i^n |y_i - a_i^L| \quad (9)$$

where y_i is the observed chronological brain age and a_i^L is the predicted brain age.

All global-context local features, F^G , serve as a single concatenated input to the MLP in the proposed model. To ensure the weights and biases adequately capture the brain’s morphological information, we want to train the GLT in the initial designed regression setup. Because the local pathway

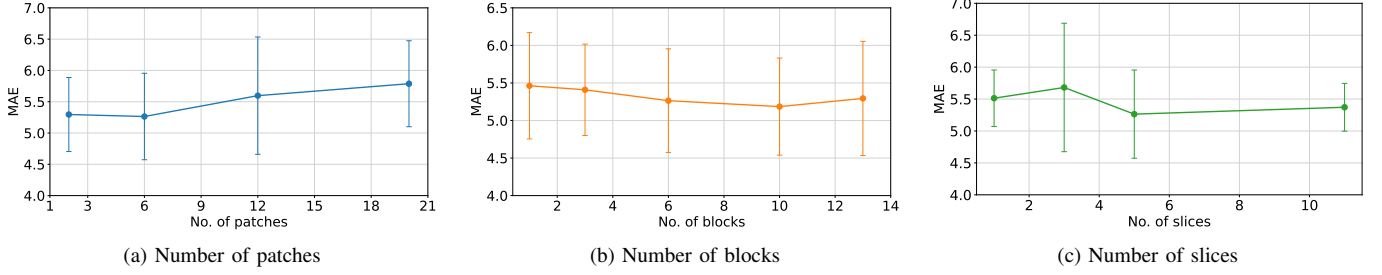


Fig. 7: MAE of GLT with different parameters: a) different patch sizes and number of patches, b) different number of blocks, c) different number of slices.

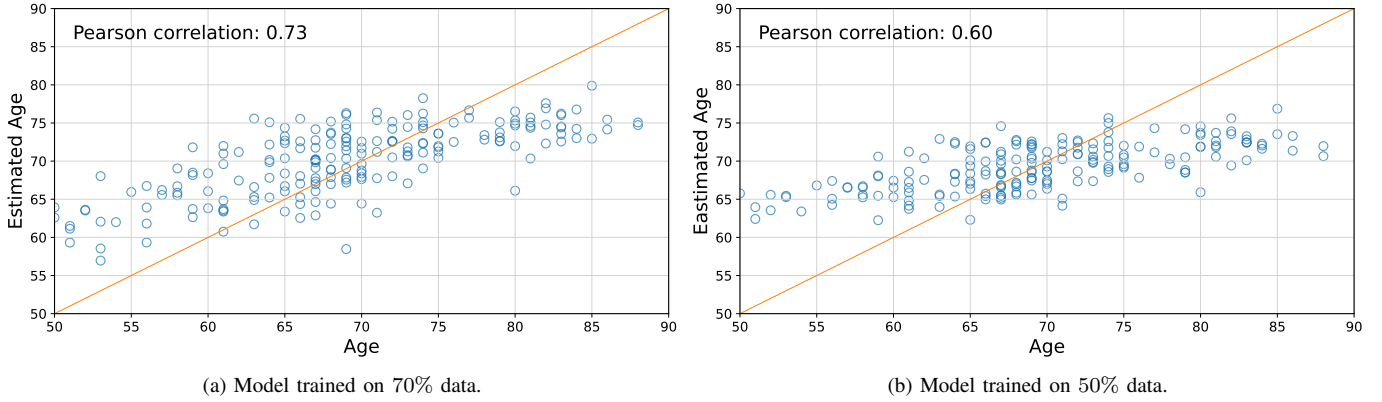


Fig. 8: The Pearson correlation on the test set for models trained on 70% and 50% of the data. The orange line is the perfect fit between the two axes.

returns predictions equal to the number of patches, our loss becomes the sum of all prediction losses from the local pathway. He et al. [6] reports the global pathway yields the highest MAE, and we therefore exclude its predictions.

B. Performance evaluation metrics

We use three metrics to evaluate brain age predictions. The MAE, which we defined in Equation 9, measures the average absolute difference between predicted and chronological ages. The model configuration results in Figure 7 report the average validation MAE across all FFCV iterations for the epoch yielding the lowest MAE.

The correlation coefficient that we report in Figure 8 is the sample correlation coefficient (ρ) between the predicted age and the chronological age, computed as follows:

$$\rho = \frac{\sum(\mathbf{x} - m_x)(\mathbf{y} - m_y)}{\sqrt{\sum(\mathbf{x} - m_x)^2 \sum(\mathbf{y} - m_y)^2}} \quad (10)$$

where m_x is the mean of the vector $\mathbf{x}$ and m_y is the mean of the vector $\mathbf{y}$ [34]. The Pearson correlation determines the strength of the linear relationship between the chronological age and the estimated age. A value between 0.5 and 1 is considered a strong positive correlation. Conversely, a value between -0.5 and -1 is a strong negative correlation.

Lastly, we report the cumulative score (CS) in Figure 9, denoting the proportion of errors within some error threshold, given as

$$CS(\alpha) = N_{e \leq \alpha} / N \times 100\% \quad (11)$$

where $N_{e \leq \alpha}$ is the number brain age predictions within the error threshold [6]. α is the error level in years.

C. Model Configuration

We describe in III-C that the optimal configuration for the GLT model is found using FFCV on NC subjects. It must be reiterated that session-level data splits during FFCV were later found to introduce bias and discredit the reported results. A single CV iteration implements 100 training epochs with an initial learning rate of 0.0001 that is reduced every 25th epoch. With those settings, we evaluate the parameters: number of patches in 7a, number of GLT blocks in 7b, and number of slices in 7c.

While investigating different patch sizes, we maintained a fixed configuration of 6 GLT blocks and 5 slices. In 7a, it becomes apparent that the minimum MAE occurs when using 6 patches. As we increase the number of patches, both training times and MAE rise, leading us to select 6 as the number of patches. We continue to use 5 slices and 6 patches while exploring the optimal number of GLT blocks. Referencing 7b, we observe that the lowest MAE is achieved with 10 GLT blocks. However, employing 10 GLT blocks almost doubles

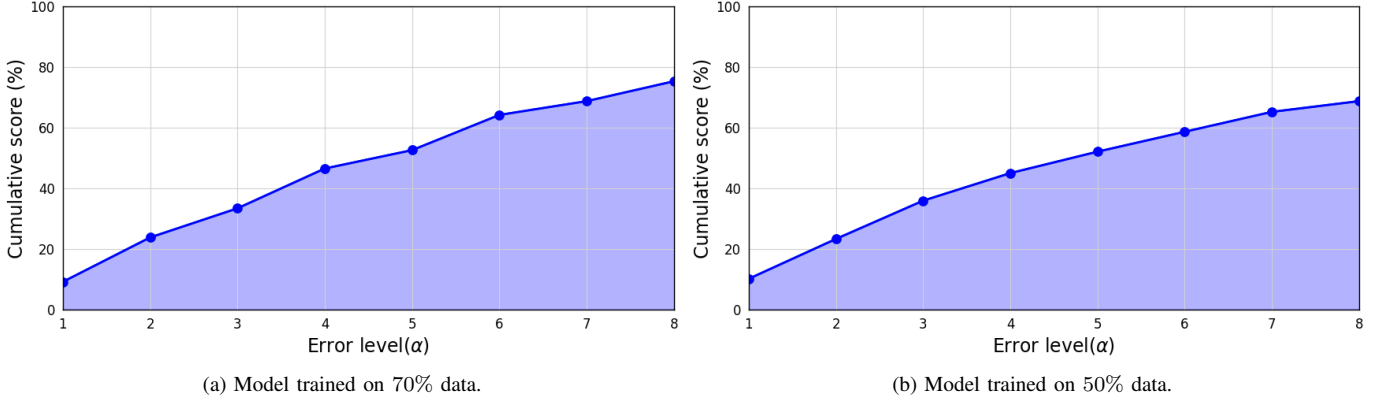


Fig. 9: The Cumulative Score (CS) on the test set, for models trained on 70% and 50% of the data, with the error level α varying from 1 to 8 years.

TABLE IV: RESULTS FOR GLT ON TEST SET

Models:	MAE	Pearson correlation	CS($\alpha = 8$)
70% model	5.34 ± 4.72	0.73	75.25%
50% model	5.76 ± 4.50	0.60	68.69%

the training time compared to 6 GLT blocks and only results in a marginal MAE improvement of less than 0.10 years. Consequently, we opted with 6 GLT blocks. Lastly, we retain the configuration of 5 slices since it yields the lowest MAE.

There are other parameters in the GLT network that we did not evaluate. One is whether to include the global pathway in the final prediction or only use the predictions of the local pathway. Additionally, one can also utilize different backbone CNNs. For those settings, we rely on the reported results by He et al. [6], and therefore exclude the global pathway and select the VGG8 CNN.

D. Regression results

To showcase the generality of the GLT, we test two models trained on 50% and 70% of the data on the hold-out set. The results can be observed in Table IV. Computing the Welch’s t-test (for unequal variances) over the two distributions, that is, the output predictions from the two models, returns a p-value of 0.001, where

$$\mathcal{H}_0 : \mu_1 = \mu_2,$$

$$\mathcal{H}_1 : \mu_1 \neq \mu_2.$$

We can subsequently have confidence that the model trained on 70% yields a better result than the model trained on 50% data. Seemingly, an MAE of 5.34 years seems quite significant when we compare it to the result from the GLT authors from Table I. Nevertheless, He et al. [6] reports higher MAE for specific age groups. He et al. [6] reports explicitly that the worst age group is 30–60, with a measured MAE of 5.12 years and an MAE of 3.93 years for subjects above 60. Considering the age groups from the OASIS-3 dataset that the models are trained on, an MAE of 5.34 years appears more reasonable. He et al. [6] reports that a fused prediction based on all planes

(sagittal, coronal, and axial) yields the lowest error. The axial plane scores the second lowest error and is chosen without further experiments due to time constraints. Additionally, the different pipelines used for data cleaning and other factors also affect the MAE we expect to see from the configuration. We, therefore, consider an MAE of 5.34 years satisfactory.

The Pearson correlation can also be observed in Table IV. As both models’ correlation coefficients are in the 0.5 – 1 interval, they possess a strong linear relationship with the chronological age.

Lastly, we measured the cumulative score for several error thresholds α . At $\alpha = 8$ years, we can capture the majority of the predictions, and the development of CS given an increasing threshold can be observed on Figure 9.

E. Model analysis

We have presented some promising results for the GLT regression modeling in V-D. The model trained on 70% of the data has a high Pearson correlation with the chronological age and promising MAE. It was mentioned in I how EAD is an essential biomarker in which a subject can exhibit symptoms of cognitive impairment caused by diseases such as AD. We want to explore whether the GLT adequately recognizes differences between NC and AD subjects regarding our regression results.

It can be observed on Figure 8 how the models tend to overestimate the age for subjects under 70 but underestimates the age for subjects over 70 years. We further investigate the model’s behavior on the test set, fit a linear model to the 70% model’s prediction, and compute the R^2 value to verify the variability in the predictions a simple fit can explain. Since we observed a high positive correlation coefficient, we expect the fitted line to represent that, which can be observed on Figure 10. The slope of the fitted line is 0.38, and it becomes apparent that the GLT model would not generalize well for subjects at the tail ends of the age distribution of the test set. Additionally, the maximum estimation age we can observe is 80. Although we can explain the majority of variance with the fit, we suspect the model is estimating the brain age around

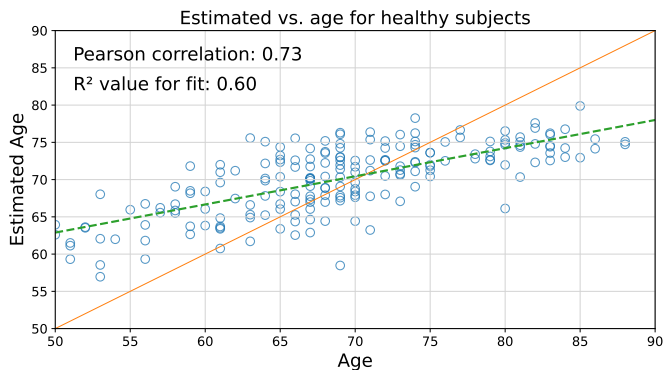


Fig. 10: The Pearson correlation on the test set for the model trained on 70% of the data. The orange line is the perfect fit between the two axes. The green line is a linear model fitted to the model’s predictions

the majority of the subject’s age distribution and neglecting the distribution’s tails.

We further investigate our suspicion by testing the 70% model on the dataset with sick subjects, which can be observed on Figure 11. As mentioned in I, individuals diagnosed with AD can be observed to have accelerated brain degeneration. The assumption is that AD patients should exhibit a higher estimated brain age than their healthy counterparts. The model produces an MAE of 6.08 years; however, it is evident from Figure 11 that we cannot explain much of the model’s predictions in the sick dataset. With a moderate Pearson correlation of 0.34 between the estimated age and the chronological age, it seems evident that the model does not produce the expected result. To say the EAD holds any information concerning the AD classification task, we would assume the slope of the fitted model to be greater than 1. For Figure 11, the slope of the fitted model is 0.15. With a small R^2 value on 0.21, the 70% model’s predictions seem on par with randomly guessing the age distribution rather than modeling the brain age. The mean age of subjects from the AD dataset is known from Table III to be 75.62 years. The average age prediction the model produces given the sick subjects is 73, 47 years, a contrasting result from the expectation. It can also be observed that no estimation is above 80, which confirms that the model does not extrapolate well.

We conclude the analysis by deciding to exclude the estimated age from GLT in the input to the MLP, as the scalar value holds no meaningful information regarding subjects diagnosed with AD. We still intend to investigate if the MLP will have any predictive power, given the feature vector from the GLT.

VI. CLASSIFICATION EXPERIMENTS AND RESULTS

The final phase of our study involves transferring our pre-trained deep network model to investigate the relationship between EAD and AD. Wen et al. [8] has highlighted challenges in comparing classification performance across studies due to variations in factors such as participant selection, image pre-processing, and validation procedures. Additionally, it has

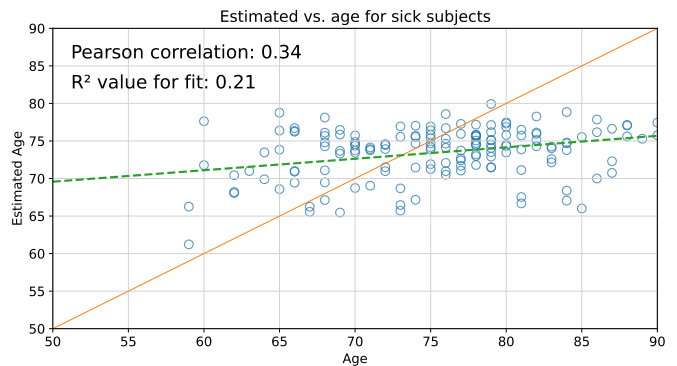


Fig. 11: The Pearson correlation on the sick subjects for the model trained on 70% of the data. The orange line is the perfect fit between the two axes. The green line is a linear model fitted to the model’s predictions

been observed in the field of health science that a substantial number of studies focusing on AD classification employing CNNs may have encountered issues related to data leakage, thereby leading to biased performance reporting [8]. To create a comparable setup to [3], we deliberately introduce data leakage and data imbalance in our study and examine its impact on the performance of our model. By doing so, we hope to gain insights into the potential consequences of these issues and their influence on our model’s performance.

A. Examining data leakage and imbalance

We conducted a series of experiments involving the transformer model with varying proportions of the dataset. When the transformer model was trained on 100% of NCs without an independent test set, it resulted in biased transfer learning. This occurs because the transformer model has been exposed to all the NC samples during the regression training.

We also performed experiments with a 70% and a 50% split of the data, ensuring the inclusion of an independent test set for AD classification. For each of these three experiments, we conducted additional experiments to investigate the impact of data imbalance. However, due to limitations in the available data, it was not possible to introduce the same degree of imbalance for each data proportion.

We considered an equal split of NC and AD samples to define a balanced dataset. We also defined mild imbalance as a scenario where the minority class constituted 32% of the data, indicating that 32% of the samples were from the AD class. Lastly, we defined moderate imbalance as a situation where the minority class accounted for 19% of the data.

The learned feature vector extracted from the GLT was utilized as input for the MLP in the context of AD diagnosis. The model was trained using BCE loss and the Adam optimizer, with an initial learning rate of 0.00001 that dynamically decreases during training.

B. Classification evaluation metrics

To evaluate the performance of the model, we employed a FFCV strategy. Accuracy, Area under the ROC Curve (ROC-AUC), Sensitivity, Specificity, and their respective standard

TABLE V: CROSS-VALIDATION RESULTS FOR TRANSFORMER TRAINED ON 100% NC

Data Imbalance Rate	Accuracy	ROC-AUC score	Sensitivity	Specificity	Best Epoch
Balanced	91.76% $\pm$ 2.06%	0.98 $\pm$ 0.01	0.91 $\pm$ 0.02	0.93 $\pm$ 0.04	95
Mildly Imbalanced	93.97% $\pm$ 1.25%	0.98 $\pm$ 0.01	0.87 $\pm$ 0.03	0.98 $\pm$ 0.01	97
Moderately Imbalanced	95.43% $\pm$ 1.31%	0.98 $\pm$ 0.01	0.80 $\pm$ 0.06	0.99 $\pm$ 0.01	97

TABLE VI: CROSS-VALIDATION RESULTS FOR TRANSFORMER TRAINED ON 50% NC

Data Imbalance Rate	Accuracy	ROC-AUC score	Sensitivity	Specificity	Best Epoch
Balanced	76.91% $\pm$ 4.02%	0.82 $\pm$ 0.04	0.81 $\pm$ 0.09	0.73 $\pm$ 0.11	3
Mildly Imbalanced	76.76% $\pm$ 2.91%	0.81 $\pm$ 0.04	0.57 $\pm$ 0.09	0.86 $\pm$ 0.04	35

TABLE VII: CROSS-VALIDATION RESULTS FOR TRANSFORMER TRAINED ON 70% NC

Data Imbalance Rate	Accuracy	ROC-AUC score	Sensitivity	Specificity	Best Epoch
Balanced	75.09% $\pm$ 5.22%	0.81 $\pm$ 0.06	0.80 $\pm$ 0.10	0.71 $\pm$ 0.03	78

deviation (SD) were computed as evaluation metrics. Accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}, \quad (12)$$

where the true positives (TPs) and true negatives (TNs) represent the number of correctly labeled AD subjects. The denominator includes the false positive (FP) and false negative (FN) labels, thus, representing the total number of subjects.

Specificity and Sensitivity are defined as

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad (13)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}. \quad (14)$$

Specificity represents how good a model is at classifying NC, whereas, Sensitivity represents how good a model is at classifying AD. Lastly, ROC-AUC is defined as

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(FPR) dFPR, \quad (15)$$

where the true positive rate (TPR), commonly referred to as Sensitivity, and the false positive rate (FPR), denoted as 1 – Sensitivity, are calculated across different threshold values, contributing to the construction of the Receiver Operating Characteristic (ROC) curve. ROC-AUC score is a comprehensive measure to evaluate the classifier’s performance, with values ranging from 0 to 1. A higher ROC-AUC score indicates superior classifier performance, while a ROC-AUC of 0.5 suggests random guessing, and a ROC-AUC of 1.0 signifies a perfect classifier. The result of the measures can be seen in Table V, Table VI, and Table VII.

C. Alzheimer’s disease diagnosis results

Table V presents the performance analysis of the transformer trained on 100% of the NC data, demonstrating superior results across all evaluated metrics. Notably, introducing data leakage and a moderately imbalanced dataset yields an accuracy of 95.43% and a ROC-AUC score of 0.98. In Table VI and Table VII, where data leakage is eliminated, the model achieves an accuracy of 76.91% and a ROC-AUC score

of 0.82 on a balanced dataset. These results indicate that the presence of both data leakage and data imbalance significantly enhances the accuracy of AD diagnosis. Additionally, the results are selected by identifying the epoch number that yielded the highest accuracy, as represented in the tables.

Furthermore, the sensitivity score consistently decreases as the data imbalance rate increases, reflecting a diminished ability to identify individuals with AD correctly. Conversely, the specificity increases with higher data imbalance rates, indicating an improved ability to classify healthy individuals accurately. These observations emphasize the impact of imbalanced data on classification performance.

VII. DISCUSSION

In the health science domain, obtaining high-quality and quantity datasets can often be challenging due to regulations and privacy concerns. Acquiring data can be a lengthy process and has been the biggest limiter in this paper. A more comprehensive range of data would have provided more options for training, testing, and validating the models. The age distributions are also a repetitive topic in this paper, as it is evident that we have not implemented enough measures to accommodate the minority representations, and they are not weighted enough in the training and validation phase. It is evident when the classifier is validated on imbalanced datasets. Simple measures for this include but are not limited to, downsampling and upweighting, where the majority class is scaled down to a more representative proportion of the minority class and then upweighted to a factor equal to the downsampling [35].

We identified a data leak issue in the project, which we did our best to accommodate. Unfortunately, due to the sheer size of the GLT network, we did not have the necessary time scope to address this issue and re-iterate the validation process. Therefore, the GLT model validation differs from the standard we have wished for. After the data leak was discovered, we implemented robust measures to ensure such a leak would not occur again. Additionally, this meant that a healthy subject found in the healthy dataset would be removed from the sick dataset if an appearance was made.

TABLE VIII: AGE DISTRIBUTION ON VARYING PROPORTIONS OF THE DATASET

	Age statistics
AD	75.52 $\pm$ 7.70
NC Balanced 1	67.06 $\pm$ 8.81
NC Mildly Imbalanced 1	67.58 $\pm$ 9.53
NC Moderately Imbalanced 1	67.96 $\pm$ 9.49
NC Balanced 05	67.97 $\pm$ 9.60
NC Mildly Imbalanced 05	68.35 $\pm$ 9.43
NC Balanced 07	68.85 $\pm$ 8.95

The proposed method we present in this paper performs competitively on AD classification. However, due to the nature of the data-driven models utilized in the method, we also have to address the resilience and soundness of the models' results. We briefly touch on the predictive capabilities of a trained GLT in our regression configuration in V-E. The GLT achieves an adequate MAE; however, it can quickly be verified that the model does not generalize well to other datasets where the age distribution does not resemble the training set. In that aspect, we cannot guarantee that the classifier has learned to segregate AD from NC based on accelerated brain degeneration. The classifier likely learned to separate based on the two different age distributions from the NC and AD datasets. This difference in the age distribution is shown in Table VIII.

VIII. CONCLUSION

In this paper, we have employed two different models, the GLT and the MLP, that can successfully segregate AD subjects from NC subjects. We present a fair measure for the balanced dataset that produces an impressive accuracy on 76.91%, comparable with state-of-the-art methods, with no data leak. A comparison can be observed in Table II. Furthermore, introducing data leakage and imbalance leads to a substantial improvement in accuracy, with an accuracy of 95.43%. The model achieves near-perfect classification of all NC subjects, as indicated by the specificity value of 0.99 presented in Table V. However, the model exhibits shortcomings in accurately classifying individuals with AD, resulting in a biased accuracy.

ACKNOWLEDGEMENTS

We leveraged the assistance of ChatGPT [36] and GitHub Copilot [37], which provided valuable code suggestions, expediting the development process and serving as a source of inspiration when encountering challenges.

This study was made possible by Aalborg University (AAU) and their cloud department, CLAAUDIA. The dataset was kindly provided by Open Access Series of Imaging Studies (OASIS). The work presented in this paper was supervised by Dalin Zhang and Ehsan Mobaraki of Aalborg University.

REFERENCES

- [1] Galluzzi, Samantha et al. "Aging." *Neurological sciences : official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology* vol. 29 Suppl 3 (2008): 296-300. <https://doi.org/10.1007/s10072-008-1002-6>
- [2] Mishra, Shiwangi et al. "A Review of Neuroimaging-Driven Brain Age Estimation for Identification of Brain Disorders and Health Conditions." *IEEE reviews in biomedical engineering* vol. 16 (2023): 371-385. <https://doi.org/10.1109/RBME.2021.3107372>
- [3] H. Cai, Y. Gao and M. Liu, "Graph Transformer Geometric Learning of Brain Networks Using Multimodal MR Images for Brain Age Estimation," in *IEEE Transactions on Medical Imaging*, vol. 42, no. 2, pp. 456-466, Feb. 2023, <https://doi.org/10.1109/TMI.2022.3222093>.
- [4] Cole, James H et al. "Brain age and other bodily 'ages': implications for neuropsychiatry." *Molecular psychiatry* vol. 24,2 (2019): 266-281. <https://doi.org/10.1038/s41380-018-0098-1>
- [5] Peters, R. "Ageing and the brain." *Postgraduate medical journal* vol. 82,964 (2006): 84-8. <https://doi.org/10.1136/pgmj.2005.036665>
- [6] S. He, E. Grant, Y. Ou, *Global-Local Transformer for Brain Age Estimation*, *IEEE Trans Med Imaging*, 2022, <https://doi.org/10.1109/TMI.2021.3108910>.
- [7] Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017), <https://doi.org/10.48550/arXiv.1706.03762>.
- [8] Junhao Wen, Elina Thibeau-Sutre, Mauricio Diaz-Melo, Jorge Samper-González, Alexandre Routier, Simona Bottani, Didier Dormont, Stanley Durrleman, Ninon Burgos, Olivier Colliot. "Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation". <https://doi.org/10.1016/j.media.2020.101694>.
- [9] Cole, James H et al. "Predicting brain age with deep learning from raw imaging data results in a reliable and heritable biomarker." *NeuroImage* vol. 163 (2017): 115-124. <https://doi.org/10.1016/j.neuroimage.2017.07.059>
- [10] Aly Valliani and Ameet Soni. 2017. Deep Residual Nets for Improved Alzheimer's Diagnosis. In *Proceedings of the 8th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB '17)*. Association for Computing Machinery, New York, NY, USA, 615. <https://doi.org/10.1145/3107411.3108224>
- [11] U. Senanayake, A. Sowmya and L. Dawes, "Deep fusion pipeline for mild cognitive impairment diagnosis," 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018), Washington, DC, USA, 2018, pp. 1394-1997, <https://doi.org/10.1109/ISBI.2018.8363832>.
- [12] Jack, Clifford R Jr et al. "The Alzheimer's Disease Neuroimaging Initiative (ADNI): MRI methods." *Journal of magnetic resonance imaging : JMIR* vol. 27,4 (2008): 685-91. <https://doi.org/10.1002/jmri.21049>
- [13] P. J. LaMontagne, T. L. S. Benzinger, J. C. Morris, S. Keefe, R. Hornbeck, C. Xiong, E. Grant, J. Hasselstab, K. Moulder, A. G. Vlassenko, M. E. Raichle, C. Cruchaga, D. Marcus, *OASIS-3: Longitudinal Neuroimaging, Clinical, and Cognitive Dataset for Normal Aging and Alzheimer Disease*, 2019.12.13.19014902, 2019, <https://doi.org/10.1101/2019.12.13.19014902>, Cold Spring Harbor Laboratory Press.
- [14] Luo W, Phung D, Tran T, Gupta S, Rana S, Karmakar C, Shilton A, Yearwood J, Dimitrova N, Ho TB, Venkatesh S, Berk M. Guidelines for Developing and Reporting Machine Learning Predictive Models in Biomedical Research: A Multidisciplinary View. *J Med Internet Res*. 2016 Dec 16;18(12):e323. <https://doi.org/10.2196/jmir.5870>. PMID: 27986644; PMCID: PMC5238707.
- [15] K. Bäckström, M. Nazari, I. Y. -H. Gu and A. S. Jakola, "An efficient 3D deep convolutional network for Alzheimer's disease diagnosis using MR images," 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018), Washington, DC, USA, 2018, pp. 149-153, <https://doi.org/10.1109/ISBI.2018.8363543>.
- [16] NrgXnat (2021, August 10), *oasis-scripts*, GitHub Repository, <https://github.com/NrgXnat/oasis-scripts>, visited on April 24rd 2023.
- [17] BIDS, *Brain Imaging Data Structure*, <https://bids.neuroimaging.io/index.html>, visited on April 23rd 2023.
- [18] K.V. Nielsen, E.M. Buchberg, F.H. Kowalski, *p6*, Github Repository, https://github.com/kent1325/2D-Brain-age-estimation_AD-classification_using_DL, visited on May 24th 2023.
- [19] European Parliament & Council of the European Union, *Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (Text with EEA relevance)*, OJ L 119, 4.5.2016, p. 1–88, visited on May 16th, from <https://gdpr-info.eu/recitals/>.
- [20] A. Routier, N. Burgos, M. Díaz, M. Bacci, S. Bottani, O. El-Rifai, S. Fontanella, P. Gori, J. Guillon, A. Guyot, R. Hassanaly, T. Jacquemont, P. Lu, A. Marcoux, T. Moreau, J. Samper-González, M. Teichmann, E. Thibeau-Sutre, G. Vaillant, J. Wen, A. Wild, M-O. Habert, S. Durrleman and O. Colliot, (2021) *Clinica: An Open-Source Software Platform for Reproducible Clinical Neuroscience Studies.*, *Frontiers in*

- Neuroinformatics, vol. 15, 2021, ISSN 1662-5196, doi: <https://doi.org/10.1109/ISBI.2018.8363543/10.3389/fninf.2021.689675>.
- [21] J. Wen, E. Thibeu-Sutre, M. Diaz-Melo, J. Samper-González, A. Routier, S. Bottani, D. Dormont, S. Durrleman, N. Burgos, O. Colliot, *Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation*, Medical Image Analysis, vol. 63, 2020, 101694, ISSN 1361-8415, <https://doi.org/10.1016/j.media.2020.101694>.
 - [22] N. J. Tustison et al., *N4ITK: Improved N3 Bias Correction*, IEEE Transactions on Medical Imaging, vol. 29, no. 6, pp. 1310-1320, June 2010, <https://doi.org/10.1109/TMI.2010.2046908>.
 - [23] SimpleITK, *N4 Bias Field Correction*, https://simpleitk.readthedocs.io/en/master/link_N4BiasFieldCorrection_docs.html, visited on April 23rd 2023.
 - [24] Juntu, Jaber, et al. *Bias Field Correction for MRI Images*, Computer Recognition Systems, Springer Berlin Heidelberg, pp. 543–51, https://doi.org/10.1007/3-540-32390-2_64.
 - [25] B.B. Avants, C.L. Epstein, M. Grossman, J.C. Gee, *Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain*, Medical Image Analysis, vol. 12, Issue 1, 2008, Pages 26-41, ISSN 1361-8415, <https://doi.org/10.1016/j.media.2007.06.004>.
 - [26] B.B. Avants, N.J. Tustison, M. Stauffer, G. Song, B. Wu, J.C. Gee, *The Insight ToolKit image registration framework*, Frontiers in Neuroinformatics, vol. 8, 2014, <https://www.frontiersin.org/articles/10.3389/fninf.2014.00044>, ISSN 1662-5196, doi: 10.3389/fninf.2014.00044.
 - [27] V.S. Fonov, A.C. Evans, R.C. McKinstry, C.R. Alml, D.L. Collins, *Unbiased nonlinear average age-appropriate brain templates from birth to adulthood*, NeuroImage, vol. 47, Supplement 1, 2009, Page S102, ISSN 1053-8119, [https://doi.org/10.1016/S1053-8119\(09\)70884-5](https://doi.org/10.1016/S1053-8119(09)70884-5).
 - [28] ClinicaDL, *Prepare-data pipeline*, <https://clinicadl.readthedocs.io/en/latest/Preprocessing/Extract/>, visited on May 15th 2023.
 - [29] E. Thibeu-Sutre, M. Díaz, R. Hassanaly, A. Routier, D. Dormont, O. Colliot, N. Burgos, *ClinicaDL: An open-source deep learning software for reproducible neuroimaging processing*, Computer Methods and Programs in Biomedicine, vol. 220, 2022, 106818, ISSN 0169-2607, <https://doi.org/10.1016/j.cmpb.2022.106818>.
 - [30] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, *PyTorch: An Imperative Style, High-Performance Deep Learning Library*, https://proceedings.neurips.cc/paper_files/paper/2019/file/bdbca288fee7f92f2bfa9f7012727740-Paper.pdf.
 - [31] V. S. Fonov, M. Dadar, The PREVENT-AD Research Group ADNI, D. L. Collins, *DARQ: Deep learning of quality control for stereotaxic registration of human brain MRI to the T1w MNI-ICBM 152 template*, NeuroImage, vol. 257, 2022, 119266, ISSN 1053-8119, doi: <https://doi.org/10.1016/j.neuroimage.2022.119266>.
 - [32] D. L. Poole, A. K. Mackworth, *Artificial Intelligence: Foundations of Computational Agents*, Cambridge University Press, 2017.
 - [33] M. A. Nielsen, *Neural Networks and Deep Learning*, Determination Press, 2015.
 - [34] SciPy community (2021), *scipy.stats.pearsonr*, SciPy v1.7.0 Reference Guide, <https://docs.scipy.org/doc/scipy/reference/generate/scipy.stats.pearsonr.html#id2>, visited on may 20th 2023.
 - [35] "Sampling and Splitting Data." Google Developers, Google, 2023, developers.google.com/machine-learning/data-prep/construct/sampling-splitting/imbalanced-data.
 - [36] ChatGPT (2022). *ChatGPT: Optimizing Language Models for Dialogue*. OpenAI. Retrieved from <https://openai.com/blog/chatgpt> Access Date 19-05-2023.
 - [37] Friedman, Nat. "Introducing GitHub Copilot: your AI pair programmer." (2021). Retrieved from <https://copilot.github.com/> Access Date 19-05-2023.